

ally, in the appendix we present alternative zero-shot prompts and task specifications (Appendix K.3 and K.1 respectively), but since these did not improve performance they were not further considered. Another limitation is the fact that a subset of the models we evaluate are behind APIs. This means models are subject to change (affecting reproducibility) and certain details about these models are unknown. This affects the group instruction-tuned at the example-level, which is the group we find outperforms all others and has the most favourable scaling properties. How do we know instruction-tuning at the example-level is the main driver behind these findings without controlled A/B testing? Unfortunately, due to the secrecy surrounding the exact implementation of these models we cannot be certain, but we can be relatively confident. We evaluated ten models across six model classes and two APIs in the group example-level instruction tuned. Within this group, models probably vary significantly in other training and architecture details (especially Cohere-command models versus OpenAI models). The most salient commonality they share with each other and none of the other models is multi-task instruction-tuning at the example level, making it likely that this is the driving factor of their performance. A further datapoint in favour of this conclusion can be seen in Figure 3 (right); base models at similar scales as Example IT models perform significantly worse. We see that Cohere-command 52B significantly outperforms Cohere-base 52B, and the only difference between those models is instruction-tuning at the example level (Cohere-command is fine-tuned from Cohere-base). In fact, Cohere-command 52B outperforms other base models more than 3 times the size by a large margin (e.g. GPT-3 175B, BLOOM-176B, OPT-175B). We are therefore confident that instruction-tuning at the example-level is important for pragmatic understanding, an insight which can guide the development of open-source models capable of pragmatic understanding. Investigating the exact effect of this type of instruction-tuning on pragmatic understanding in a controlled setting is an interesting future work direction (e.g. by isolating the effect of data diversity from instructions). Another limitation is that some evaluations are subject to API stochasticity, which we address in Appendix K.5. After running the zero-shot experiment ten times through each API we conclude there is some stochasticity, but it is too small to impact our conclusions. We publish exact timestamps at which we queried APIs in Appendix L. Further, a downside of doing a comprehensive analysis on many models is compute costs. In Appendix M we publish a list of exact compute used (time and hardware), as well as estimated carbon emissions for each of the models that are not behind an API. Finally, the likelihood ranking approach we take limits our study to implicatures with clear alternative. However, implicatures in natural language can entail more complex propositions. For example, imagine Esther now asking “Can I use your stapler?” and Juan responding “Here’s the key to my office.”. Juan is implicating that (1) Esther can use the stapler, (2) the stapler is located in the office, and (3) the office is currently locked. This leaves ample room for the design of benchmarks with implicatures entailing multiple non-binary propositions.

6 Conclusion

LLMs have made remarkable progress on fluency and coherence in recent years. We argue however that a central aspect of language understanding is missing from evaluations. To understand language means to understand its pragmatics: its usage in a context that incorporates commonsense understanding, goals, objectives, and so on. We design a protocol that evaluates LLMs on binary implicature resolution and establish a significant gap with human understanding for SotA LLMs in three categories; large-scale pre-trained models, models fine-tuned on conversations, and models fine-tuned with benchmark-level instructions. By contrast, we find that models fine-tuned on example-level instructions perform significantly better. This group also exhibits the best correlation between accuracy and model size. Scaling analysis shows that for some large-scale pre-trained models accuracy also positively correlates with model size, but the best model in this group would need at least five times more parameters to reach similar performance. From these results, we conclude that instruction-tuning at the example level is important for pragmatic understanding. We hypothesise that there is something about the multi-task data diversity obtained from example-level instructions (i.e. each example a new task) that makes pragmatic understanding appear at smaller scale.

Acknowledgements

We would like to thank members of the UCL DARK lab, members of the UCL NLP group, Pontus Stenetorp, Sebastian Borgeaud, Philipp Jettkant, Robert Kirk, and Max Bartolo for fruitful discussions and comments on an earlier draft of this paper. We would also like to thank the anonymous reviewers for engaging actively in discussion and providing feedback that has improved this work. This work was supported by the EPSRC Grant EP/S021566/1 and UCL International Scholar Award for Doctoral Training Centres.

References

- Adolphs, L., Börschinger, B., Buck, C., Huebscher, M. C., Ciaramita, M., Espeholt, L., Hofmann, T., Kilcher, Y., Rothe, S., Sessa, P. G., and Sestorain, L. (2022). Boosting search engines with interactive agents. *Transactions on Machine Learning Research*.
- American Psychiatric Association, A. P. A. (2013). *Diagnostic and statistical manual of mental disorders : DSM-5*. American Psychiatric Association Arlington, VA, 5th ed. edition.
- Askell, A., Bai, Y., Chen, A., Drain, D., Ganguli, D., Henighan, T., Jones, A., Joseph, N., Mann, B., DasSarma, N., Elhage, N., Hatfield-Dodds, Z., Hernandez, D., Kernion, J., Ndousse, K., Olsson, C., Amodei, D., Brown, T. B., Clark, J., McCandlish, S., Olah, C., and Kaplan, J. (2021). A general language assistant as a laboratory for alignment. *CoRR*, abs/2112.00861.
- Austin, J., Odena, A., Nye, M., Bosma, M., Michalewski, H., Dohan, D., Jiang, E., Cai, C., Terry, M., Le, Q., et al. (2021). Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*.
- Bach, K. (1999). The myth of conventional implicature. *Linguistics and Philosophy*, 22(4):327–366.
- Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., DasSarma, N., Drain, D., Fort, S., Ganguli, D., Henighan, T., et al. (2022). Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*.
- Bender, E. M., Gebru, T., McMillan-Major, A., and Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pages 610–623.
- Biderman, S. and Raff, E. (2022). Fooling moss detection with pretrained language models. *arXiv preprint arXiv:2201.07406*.
- BigScience (2022). Bigscience language open-science open-access multilingual (bloom) language model.
- Black, S., Biderman, S., Hallahan, E., Anthony, Q., Gao, L., Golding, L., He, H., Leahy, C., McDonnell, K., Phang, J., Pieler, M., Prashanth, U. S., Purohit, S., Reynolds, L., Tow, J., Wang, B., and Weinbach, S. (2022). GPT-NeoX-20B: An open-source autoregressive language model. In *Proceedings of the ACL Workshop on Challenges & Perspectives in Creating Large Language Models*.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., McCandlish, S., Radford, A., Sutskever, I., and Amodei, D. (2020). Language models are few-shot learners. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc.
- Chung, H. W., Hou, L., Longpre, S., Zoph, B., Tay, Y., Fedus, W., Li, E., Wang, X., Dehghani, M., Brahma, S., Webson, A., Gu, S. S., Dai, Z., Suzgun, M., Chen, X., Chowdhery, A., Narang, S., Mishra, G., Yu, A., Zhao, V., Huang, Y., Dai, A., Yu, H., Petrov, S., Chi, E. H., Dean, J., Devlin, J., Roberts, A., Zhou, D., Le, Q. V., and Wei, J. (2022). Scaling instruction-finetuned language models.
- Cianflone, A., Feng, Y., Kabbara, J., and Cheung, J. C. K. (2018). Let’s do it “again”: A first computational approach to detecting adverbial presupposition triggers. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2747–2755, Melbourne, Australia. Association for Computational Linguistics.
- Davis, W. (2019). Implicature. In Zalta, E. N., editor, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, fall 2019 edition.

- Davis, W. A. (1998). *Implicature : intention, convention, and principle in the failure of Gricean theory* / Wayne A. Davis. Cambridge studies in philosophy. Cambridge University Press, Cambridge England ; New York.
- Devlin, J., Chang, M., Lee, K., and Toutanova, K. (2018). BERT: pre-training of deep bidirectional transformers for language understanding. *CoRR*, abs/1810.04805.
- Efrat, A. and Levy, O. (2020). The turking test: Can language models understand instructions? *CoRR*, abs/2010.11982.
- Frank, M. C. and Goodman, N. D. (2012). Predicting pragmatic reasoning in language games. *Science*, 336:998 – 998.
- Gao, L., Tow, J., Biderman, S., Black, S., DiPofi, A., Foster, C., Golding, L., Hsu, J., McDonell, K., Muennighoff, N., Phang, J., Reynolds, L., Tang, E., Thite, A., Wang, B., Wang, K., and Zou, A. (2021). A framework for few-shot language model evaluation.
- Gehman, S., Gururangan, S., Sap, M., Choi, Y., and Smith, N. A. (2020). RealToxicityPrompts: Evaluating neural toxic degeneration in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 3356–3369, Online. Association for Computational Linguistics.
- George, E. J. and Mamidi, R. (2020). Conversational implicatures in english dialogue: Annotated dataset. *Procedia Computer Science*, 171:2316–2323. <https://doi.org/10.1016/j.procs.2020.04.251>.
- Glaese, A., McAleese, N., Trebacz, M., Aslanides, J., Firoiu, V., Ewalds, T., Rauh, M., Weidinger, L., Chadwick, M., Thacker, P., Campbell-Gillingham, L., Uesato, J., Huang, P.-S., Comanescu, R., Yang, F., See, A., Dathathri, S., Greig, R., Chen, C., Fritz, D., Sanchez Elias, J., Green, R., Mokrá, S., Fernando, N., Wu, B., Foley, R., Young, S., Gabriel, I., Isaac, W., Mellor, J., Hassabis, D., Kavukcuoglu, K., Hendricks, L. A., and Irving, G. (2022). Improving alignment of dialogue agents via targeted human judgements.
- Goodman, N. and Frank, M. (2016). Pragmatic language interpretation as probabilistic inference. *Trends in Cognitive Sciences*, 20.
- Green, G. (1996). *Pragmatics and Natural Language Understanding*. Tutorial essays in cognitive science. Erlbaum.
- Greene, S. and Resnik, P. (2009). More than words: Syntactic packaging and implicit sentiment. In *Proceedings of Human Language Technologies: The 2009 Annual Conference of the North American Chapter of the Association for Computational Linguistics*, pages 503–511, Boulder, Colorado. Association for Computational Linguistics.
- Grice, H. P. (1975). Logic and conversation. In Cole, P. and Morgan, J. L., editors, *Syntax and Semantics: Vol. 3: Speech Acts*, pages 41–58. Academic Press, New York.
- Hosseini, M. J., Radlinski, F., Pareti, S., and Louis, A. (2023). Resolving indirect referring expressions for entity selection. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12313–12335, Toronto, Canada. Association for Computational Linguistics.
- Huang, W., Abbeel, P., Pathak, D., and Mordatch, I. (2022). Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. *arXiv preprint arXiv:2201.07207*.
- Huang, Y. (2017). *The Oxford Handbook of Pragmatics*. Oxford handbooks in linguistics. Oxford University Press.
- Jeretic, P., Warstadt, A., Bhooshan, S., and Williams, A. (2020). Are natural language inference models IMPPRESSive? Learning IMPLICature and PRESupposition. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8690–8705, Online. Association for Computational Linguistics.
- Kasirzadeh, A. and Gabriel, I. (2022). In conversation with artificial intelligence: aligning language models with human values.